
When can linear kinetic theory be trusted for magnetized cross-beam energy transfer? A particle-in-cell validity map

MagCBET Regime-Validity Study

(code, input decks, and logs accompany this manuscript)

Abstract

Cross-beam energy transfer (CBET) redistributes laser power inside inertial confinement fusion (ICF) targets, and modern implosions are increasingly magnetized, by external coils or self-generated fields, at strengths where the electron gyrofrequency is comparable to the frequencies of the mediating plasma waves. A linearized Vlasov description of magnetized CBET has recently become available and is attractive for design work because it is orders of magnitude cheaper than kinetic simulation; published particle-in-cell (PIC) tests validate it qualitatively but report unexplained quantitative differences. We map *where* the linear kinetic description remains quantitatively predictive and where it breaks, over magnetic-field strength (0–100 MG), field orientation, electron temperature (0.4–1.2 keV), laser intensity ($0.25\text{--}4 \times 10^{15} \text{ W/cm}^2$), and beam polarization, using the published processed PIC data of Shi and Moody together with new reduced-scale EPOCH simulations at conditions the original study did not cover. With a pre-registered residual-tolerance rule, a single drive-normalization anchor per plasma family, and explicit finite-duration and refraction corrections, we find that the linear model reproduces the full magnetized response of the reference carbon plasma—including the suppression of parallel-polarized transfer, the activation of cross-polarized transfer, and their crossing near 50 MG—across all field orientations up to at least 50–80 MG. The clearest failures concentrate in (i) the cross-polarized channel when the beat wavevector is nearly perpendicular to $\mathbf{B}$, where the model overpredicts the activated transfer and where variant comparison excludes magnetized-ion-mode structure as the cause; (ii) laser intensity above $1 \times 10^{15} \text{ W/cm}^2$, where transfer saturates below the linear scaling while the measurement’s own prerequisites (quiet null channel, undepleted pump) fail alongside the theory; and (iii) 100 MG fields ($\omega_{ce}/\omega_0 = 0.33$), where the theory *underpredicts* by the factor expected from the magnetized-quiver drive enhancement its averaged ponderomotive approximation drops. A compactness analysis shows that a single geometric variable, the angle between the beat wavevector and $\mathbf{B}$, together with the drive strength, carries most of the validity signal. We provide per-condition statistical support, mechanism attributions with their evidential status, and practical guidance for when ICF modeling can rely on linear magnetized-CBET predictions.

1 Introduction

Cross-beam energy transfer is the resonant exchange of energy between two crossing laser beams mediated by a low-frequency plasma wave, usually an ion-acoustic wave (IAW) quasimode driven at the beat of the two beams [1, 2]. In indirect-drive ICF it is exploited deliberately: wavelength detuning between beam cones redistributes power inside the hohlraum and controls implosion symmetry [3–5]. In direct drive it is a parasitic loss that redirects incoming light into unabsorbed directions [6–8].

In both settings, design codes rely on *linear, steady-state* models of the transfer, whose accuracy conditions have therefore received sustained attention—polarization effects and saturation [9, 10], nonlinear and kinetic corrections [11–13], and beam-smoothing effects [14].

Magnetization adds a new dimension to this question. Imposed fields of tens of tesla, compressed to MG–tens-of-MG levels, raise fusion performance and are now routine on OMEGA and NIF experiments [15–20]; comparable fields are generated spontaneously near laser entrance holes and in imploded cores [21, 22]. At MG strengths the electron cyclotron frequency ω_{ce} becomes comparable to IAW frequencies, and at tens of MG to electron-plasma-wave frequencies, so the fields modify laser-plasma instabilities directly rather than only through altered plasma conditions [23–27]. For CBET specifically, a linearized Vlasov theory of magnetized transfer (MagCBET) was recently derived by Moody and Shi [28] from an approximate magnetized ponderomotive drive, predicting reduced transfer between parallel-polarized beams, activated transfer between cross-polarized beams, and additional resonances at magnetized modes (ion Bernstein, lower hybrid, fast Alfvén). A companion 2D-3V particle-in-cell (PIC) study by Shi and Moody [29] confirmed the qualitative predictions for IAW-mediated transfer in a linear, CBET-dominant regime, while noting that “quantitative differences remain, suggesting additional physics beyond our Vlasov theory.”

For practitioners the operative question is sharper than qualitative agreement: *in which part of parameter space can the fast linear kinetic model be used with quantitative confidence, and where must one fall back to expensive kinetic simulation?* This paper answers that question with a validity map. We re-evaluate a reconstructed linear kinetic baseline against the complete processed PIC data set released with Ref. [29]—130 conditions (124 magnetized) spanning $|B| = 0.5\text{--}80$ MG, three field orientations, two polarization channels, and three plasma families—and extend the map with new reduced-scale EPOCH simulations along two directions the original study did not explore: laser intensity ($0.25\text{--}4 \times 10^{15}$ W/cm², probing the linearity assumption itself) and field strength up to 100 MG ($\omega_{ce}/\omega_0 = 0.33$, probing the theory’s drive approximation), plus a carbon temperature block that deconfounds electron temperature from plasma species. Each condition is classified *holds / marginal / breaks* by a residual-tolerance rule fixed before any comparison was made, with statistical support carried per condition, and each contiguous failure region receives a mechanism attribution tested against a pre-registered list of candidate failure modes—distinguishing genuine failures of the theory’s physical assumptions from failures of the *comparison’s* prerequisites (finite measurement duration, background losses, resolution artifacts, pump depletion).

Three results stand out. First, the linear kinetic description is *robustly quantitative* for the reference carbon conditions: with a single drive-normalization anchor fixed at $B = 0$, it tracks the measured transfer in both polarization channels, at all three orientations, from 0.5 to at least 50–80 MG—a considerably stronger statement than the published qualitative agreement. Second, its clearest in-domain failure is geometric: when the beat wavevector is nearly perpendicular to B , the activated cross-polarized transfer is overpredicted by a factor of several; comparing model variants shows the magnetized-ion mode structure contributes essentially nothing to this channel, isolating the failure in the approximate drive/coupling structure or in the electromagnetic part of the beat response that the electrostatic treatment omits. Third, the new-condition blocks bracket the linear window on two more axes: the intensity scan holds perfectly from 0.25 through 1×10^{15} W/cm² and then collapses between 1 and 2×10^{15} —with the measurement’s own prerequisites (quiet cross-polarized null channel, undepleted pump, forward-flowing energy) failing together with the theory’s linearity—while at 100 MG ($\omega_{ce}/\omega_0 = 0.33$) the model *underpredicts* by almost exactly the magnetized-quiver factor its averaged-frequency drive drops, turning the theory’s own stated validity window $\omega_{ce} \ll \omega_0$ into a measured, mechanism-attributed boundary. A compactness analysis makes the practical rule simple: the beat-field angle and the drive strength carry most of the signal; field strength alone carries surprisingly little until the drive window itself is approached.

Section 2 describes the reconstructed baseline and the measurement model; Sec. 3 the PIC data, the new simulations, and the pre-registered comparison rule; Sec. 4 the validity map; Sec. 5 the mechanism analysis; Sec. 6 the implications for ICF practice, limitations, and conclusions.

2 Linear kinetic baseline and measurement model

2.1 Configuration

We adopt the configuration of Ref. [29] throughout. A pump and a seed beam of vacuum wavelength $\lambda_0 = 351$ nm and $\lambda_1 = \lambda_0 + \Delta\lambda$ cross at angle α in the x - y plane of a uniform plasma embedded in a uniform field $\mathbf{B}$ specified by spherical angles (θ_B, ϕ_B) ; the seed propagates along $\hat{x}$. The carbon reference family has $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$ ($n_e/n_c = 0.039$), $T_e = 4T_i = 0.8 \text{ keV}$, $\alpha = 30^\circ$, $\theta_B = 90^\circ$ (field in the crossing plane), and pump and seed intensities $I_p = I_s = 0.5 \times 10^{15} \text{ W/cm}^2$; the two hydrogen families have $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i \in \{0.4, 1.2\} \text{ keV}$, $\alpha = 24^\circ$, oblique field $(\theta_B, \phi_B) = (45^\circ, 15^\circ)$, and $I = 1 \times 10^{15} \text{ W/cm}^2$. Two polarization channels are measured: z - z (pump and seed both initially polarized along $\hat{z}$) and z - y (seed initially along $\hat{y}$). The natural geometric control parameter is the angle ξ between the beat wavevector $\Delta\mathbf{k} = \mathbf{k}_0 - \mathbf{k}_1$ and $\mathbf{B}$; for the carbon geometry $\cos \xi = |\sin(\phi_B + \alpha_p/2)|$, so $\phi_B = 75^\circ, 30^\circ, -13^\circ$ correspond to $\xi \approx 0^\circ, 45^\circ, 88^\circ$ (here α_p is the in-plasma crossing angle, below).

2.2 Reconstructed linear kinetic model

The full text of the MagCBET derivation is given in Ref. [28]; since our purpose is to test the class of linearized Vlasov predictions rather than one implementation, we built and document an independent reconstruction with the same stated structure: a fluid (averaged-frequency) ponderomotive drive without magnetization corrections, a linear ion response without ion ponderomotive terms, and a kinetic plasma response evaluated along unperturbed orbits. Concretely:

Electromagnetic eigenmodes. The pump and seed are cold-plasma magnetized eigenmodes (the X/R-like and O/L-like branches) at their respective propagation angles to $\mathbf{B}$, obtained from the Stix dispersion relation; at our conditions thermal corrections to the EM branches are $O(v_{te}^2/c^2) \sim 10^{-3}$ and are neglected. Because the pump transits the narrow simulation domain in $\lesssim 20\lambda_0$, far less than an X-O beat length even at 100 MG, the pump polarization at the interaction region remains frozen at its boundary value $\hat{z}$; the infinite-plasma alternative (incoherent pump-eigenmode decomposition) is retained as a sensitivity variant.

Beat-wave response. The beat $(\Delta\omega, \Delta\mathbf{k})$, $\Delta\omega = \omega_0 - \omega_1$, drives an electrostatic response with the standard kinetic coupling factor

$$K(\Delta\omega, \Delta\mathbf{k}) = \frac{\chi_e (1 + \chi_i)}{1 + \chi_e + \chi_i}, \quad (1)$$

where χ_s are the susceptibilities of Maxwellian species in a uniform field, evaluated with the full Bessel-harmonic sums,

$$\chi_s = \frac{1}{k^2 \lambda_{Ds}^2} \left[1 + \zeta_0 \sum_{n=-\infty}^{\infty} e^{-b_s} I_n(b_s) Z(\zeta_n) \right], \quad \zeta_n = \frac{\Delta\omega - n\Omega_s}{\sqrt{2} |k_{\parallel}| v_{ts}}, \quad b_s = (k_{\perp} v_{ts} / \Omega_s)^2, \quad (2)$$

with Z the plasma dispersion function, $k_{\parallel}, k_{\perp}$ the beat wavevector components along and across $\mathbf{B}$, and Ω_s the cyclotron frequency. Equation (2) contains the magnetized IAW, its reduced/enhanced Landau damping at oblique angles, and the ion-Bernstein and lower-hybrid structure near perpendicular propagation. Three response variants are pre-declared and evaluated at every condition: the full magnetized response (both species magnetized); a *single-broad-IAW reduction* with unmagnetized ions (magnetized electrons retained), representing the “one strongly damped acoustic resonance” picture; and a null model with both species unmagnetized so that magnetization enters only through the EM eigenmodes. Differences among the variants localize which ingredient carries a given prediction. Known reconstruction limits, stated in advance: the beat response is electrostatic, so the fast-Alfvén/magnetosonic (electromagnetic) channel of Ref. [28] is not represented, and no magnetized corrections to the ponderomotive drive are included (the latter is also an approximation of the parent theory, valid in the window $\Delta\omega \ll \omega_{ce} \ll \omega_0$).

Drive and two-mode transfer. With quiver velocity $v_0 = eE_0/m_e\omega_0$ and the unmagnetized-form beat ponderomotive potential, the coupled seed eigenmode amplitudes $B_j(x)$ obey, in steady state,

$$\frac{dB}{dx} = \left[i \text{diag}(k_1, k_2) + \mathbf{M} \right] \mathbf{B}, \quad \mathbf{M}_{jm} = iP_j P_m^* \gamma_m, \quad \gamma_m = \frac{|\Delta\mathbf{k}_m|^2 v_0^2}{8c^2 k_1} K^*(\Delta\omega, \Delta\mathbf{k}_m), \quad (3)$$

where $P_j = \hat{e}_{\text{pump}} \cdot \hat{e}_j^*$ projects the (frozen) pump polarization on seed eigenmode j and $k_{1,2}$ are the seed eigenmode wavenumbers. The rank-one form of M encodes the anisotropy emphasized in unmagnetized polarization studies [9]: only the seed component parallel to the pump-projected polarization is amplified. It reproduces exactly the $B = 0$ null for orthogonal polarizations (degenerate eigenmodes), and reduces to independent exponential growth of the two eigenmodes when Faraday dephasing $(k_1 - k_2)L_x \gg 1$. The scalar prefactor was verified against the standard unmagnetized kinetic CBET formula [2, 9] to within a $\lesssim 8\%$ convention factor. Predicted observables are formed exactly as the PIC pipeline measures them: $g_j = \ln[\langle |B_j(x)|^2 \rangle_x / |B_j(0)|^2]$ and the corresponding total, i.e., the domain-averaged energy amplification of each eigenmode.

2.3 Finite-duration and refraction corrections

Two properties of the simulations (and of any ps-scale experiment) must be represented before a quantitative comparison is meaningful; both are computed from the setup geometry with no adjustable parameters, and both are B -independent, so they cannot manufacture a magnetization trend.

First, the pump enters through a boundary antenna that fixes the vacuum tangential phase, so the in-plasma pump wavevector follows from phase matching: at carbon conditions the in-plasma crossing angle is 27.95° rather than 30° , giving $|\Delta k| = 8.48 \times 10^6 \text{ m}^{-1}$ and moving the IAW resonance from $\Delta\lambda \simeq 1.23 \text{ \AA}$ (vacuum-angle estimate) to $\simeq 1.15 \text{ \AA}$.

Second, the transfer is measured 2.4–3.0 ps after the pump arrives. For the carbon plasma ($ZT_e/T_i = 24$) the IAW is *weakly* damped, $\nu \sim 2 \times 10^{10} \text{ s}^{-1}$, so the quasimode cannot ring up: $\nu T \ll 1$. The driven response accumulated over a drive of duration T is the steady response convolved, in beat frequency, with the Fejér (sinc^2) kernel of width $2\pi/T$ (bandwidth-limited by the ~ 0.6 ps pump fill time). We therefore evaluate K in Eq. (3) as this finite-duration average, at the two measurement times used by the analysis pipeline, and average the resulting gains, mirroring the measurement protocol. The effect is dramatic at $B = 0$: the ideal steady-state response is a narrow spike (peak $g \simeq 2.7$, width $\sim 0.05 \text{ \AA}$) incompatible with the broad measured curve, while the finite-duration response has the measured width (Fig. 2). The steady-state (kernel-free) evaluation is retained as a declared variant; conditions whose classification depends on it are, by construction, conditions where the steady-state assumption of the theory is the operative physics, and are flagged as such in the mechanism analysis. This transient discrimination against narrow resonances connects directly to the sub-ns CBET evolution reported in unmagnetized studies [13].

2.4 Anchoring

The reconstruction underpredicts the absolute measured gains by a factor $s = 2.66_{-0.19}^{+0.18}$ (carbon) and $s = 1.71 \pm 0.05$ (hydrogen)—an $O(1)$ normalization consistent with the crudeness of the transient-response model and unmodeled two-dimensional speckle/edge effects. Because the object under test is the *magnetization, geometry, polarization, temperature, and intensity response* of the theory rather than one overall coupling constant, the pre-registered protocol fixes a single multiplicative drive scale per plasma family at that family’s unmagnetized z - z reference condition and applies it unchanged everywhere else—never re-fit per field strength, orientation, intensity, or temperature (the hydrogen scale is shared by both temperatures so the temperature trend remains a genuine test; the same applies to the new-run carbon block). Unanchored residuals are reported alongside. All anchor values, with the propagation of the anchor-point uncertainty into every prediction, are in the analysis logs.

3 Simulation data and comparison protocol

3.1 Published processed PIC data

The validation data of Ref. [29] (released on Zenodo [38]) provide measured energy gains $g = \ln[U(t)/U(t_0)]$ of the Fourier-filtered seed—total and, for carbon, resolved into the two EM eigenmodes—with uncertainties from the spread over two measurement times and three transverse integration fractions. We use the files as released: 130 conditions in total, comprising carbon scans at fixed $\Delta\lambda = 0.9 \text{ \AA}$ over $B = 0$ –64 MG (to 80 MG for one channel) at $\phi_B \in \{75^\circ, 30^\circ, -13^\circ\}$ for both polarization channels, plus full gain curves $g(\Delta\lambda)$ at selected fields, and hydrogen gain curves over $B = 0$ –30 MG ($T_e = 0.4 \text{ keV}$) and 0–20 MG (1.2 keV). Hydrogen conditions are compared at

the detuning grid point nearest the computed unmagnetized IAW peak (1.0 Å and 1.9 Å respectively; rule fixed before any residual was computed). Measured gains contain a B -independent background offset from side- and back-scatter losses; following the original protocol it is measured on the z - y , $B = 0$ channel of each configuration and subtracted ($g_0 = -0.234$ for the carbon Fourier-method data; the real-space eigenmode method carries its own offset -0.393 , consistent with the value ≈ -0.4 quoted in Ref. [29]; each method is corrected with its own offset and the residual cross-method systematic, 0.08, is propagated where only one method exists). A systematic $\sigma_{g_0} = 0.078$ for the field dependence of this offset—estimated from the flatness of the near-perpendicular z - y channel at moderate fields—is added in quadrature to every condition.

3.2 New reduced-scale EPOCH simulations

We extend the map with 2D-3V EPOCH [37] simulations using the original study’s own input deck, changing only the physics parameters under study and the numerical resolution: 10 cells per λ_0 and 5 particles per cell instead of 20/20—exactly the reduction the original authors used for their hydrogen scans—with the domain ($500\lambda_0 \times 10\lambda_0$), boundary conditions, timing (measurements at $t_1 \approx 3.0$ ps, $t_2 \approx 3.6$ ps against the $t_0 \approx 0.61$ ps baseline), and analysis protocol unchanged. The gain-extraction pipeline is a line-by-line port of the authors’ released MATLAB analysis [39] (Fourier band-pass seed isolation, eigenmode decomposition with the Faraday phase correction, the same time/transverse-fraction uncertainty spread), verified against synthetic two-mode fields with prescribed gains and against the geometry unit tests shipped with that code. The reduced configuration was validated, before interpretation of any new condition, against the paper’s full-resolution carbon results at $\phi_B = 30^\circ$, $B \in \{0, 10, 50\}$ MG in both channels, with a pre-stated tolerance (Sec. 4.4); the RNG-seed sensitivity of the reduced configuration was measured with three-seed ensembles at two conditions, and each new configuration measures its own background offset on a z - y , $B = 0$ companion run. The new-condition blocks are: a joint pump-and-seed intensity scan $I \in \{0.25, 1, 2, 4\} \times 10^{15}$ W/cm² at $B \in \{0, 10\}$ MG; a 100 MG block ($\omega_{ce}/\omega_0 = 0.33$) at all three carbon orientations and both polarizations; and a carbon temperature block $T_e = 4T_i \in \{0.4, 1.2\}$ keV at $B \in \{0, 10\}$ MG that deconfounds temperature from species (the published hydrogen scans change species, geometry, intensity, and resolution together with T_e). Every run’s input deck, seed, raw field outputs, health diagnostics (pump-window energy budget, envelope shape, inter-time drift), and processed gains are logged.

3.3 Pre-registered comparison rule

All comparisons use the residual $\delta = |g_{\text{lin}} - g_{\text{PIC}}|$ at the fixed detuning of each family, with $\sigma^2 = \sigma_{\text{PIC}}^2 + \sigma_{g_0}^2 + \sigma_{\text{seed}}^2 + \sigma_{\text{res}}^2 + \sigma_{\text{method}}^2 + \sigma_{\text{anchor}}^2$ as applicable, and the tolerance

$$\tau = \max(0.30 |g_{\text{PIC}}|, 2\sigma, 0.15) : \quad \text{holds if } \delta \leq \tau, \quad \text{breaks if } \delta \geq 2\tau, \quad \text{marginal otherwise.} \quad (4)$$

The 30% relative tolerance reflects practical accuracy expectations for CBET modeling in design work; the 2σ term respects the statistical support of each condition; the absolute floor equals the typical published error bar so that near-null channels are not spuriously failed. Conditions where both model and measurement are consistent with zero are tagged as “both-null” (agreement, but weak evidence). The rule, the candidate mechanism list, the new-run matrix, and the reduced-scale configuration were fixed and hash-logged *before* any residual was computed; a 3×3 sensitivity grid over the relative tolerance (0.2–0.4) and floor (0.10–0.25) was declared at the same time, and classification changes under it are reported rather than explored further. Mechanism attributions follow pre-stated diagnostics separating two categories: failures of the *theory*’s assumptions (drive window, linear ion response, single-mode mediation along unperturbed orbits) versus failures of *comparison* prerequisites (steady state, undepleted pump, negligible secondary scatter, adequate resolution); the one exploratory refinement made after seeing residuals (an exclusion-based decision branch combining pre-stated variant diagnostics) is disclosed in the analysis log and below.

4 The validity map

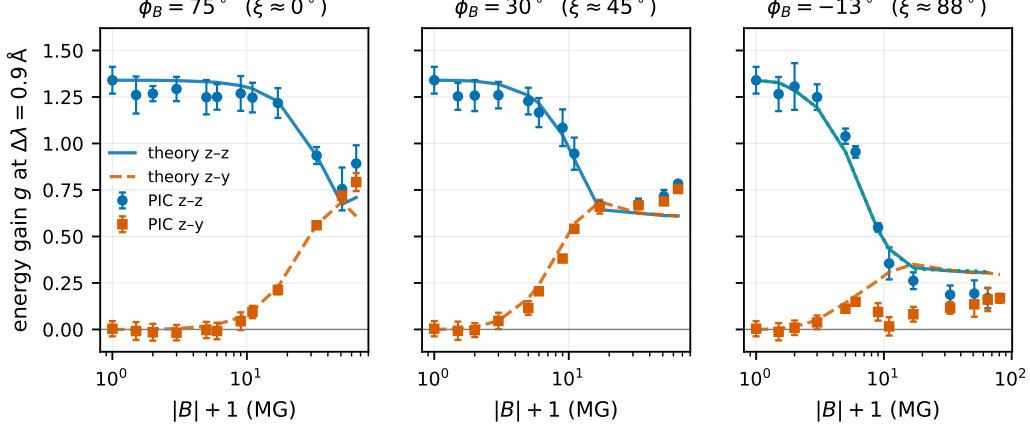


Figure 1: Carbon reference plasma: measured energy gain at $\Delta\lambda = 0.9 \text{ \AA}$ versus field strength (symbols, with the published uncertainty spread; background offset removed) against the anchored linear kinetic model (lines), for the three field orientations and both polarization channels. The single anchor is the $B = 0$ z - z point of the left panel. The dotted curve in the right panel is the single-broad-IAW variant (unmagnetized ions), indistinguishable from the full response for the z - z channel. The systematic overprediction of the near-perpendicular z - y channel (right panel, dashed) is the clearest in-domain theory failure.

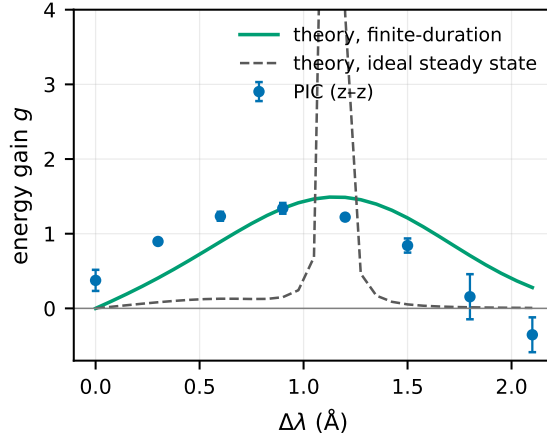


Figure 2: Unmagnetized carbon gain curve: measurement (symbols) against the anchored linear model with the finite-duration response (solid) and the ideal steady-state response (dashed). The weakly damped IAW resonance cannot reach steady state within the 2.4–3.0 ps drive, and the finite-duration kernel reproduces the measured width; a residual ~ 0.2 – $0.3 \text{ \AA}$ skew of the measured curve toward small detuning remains (Sec. 6.3).

4.1 Reference carbon conditions: quantitative validity to tens of MG

Figure 1 overlays the anchored model on the full carbon field scans. With one number fixed at $B = 0$, the linear kinetic model reproduces, within the published uncertainties, the flat-then-falling z - z response and its orientation ordering (near-parallel flat to ~ 10 MG; near-perpendicular falling from a few MG and saturating near a third of the unmagnetized value by 10 MG); the activation and growth of the z - y channel at $\xi \approx 0^\circ$ and 45° ; and the crossing of the two channels near 50 MG at about half the unmagnetized gain, exactly where the original qualitative study located it. Under the pre-registered rule (Eq. 4), 84 of the 87 carbon full-resolution conditions hold; the three exceptions are marginal points in the near-perpendicular z - y channel at 10–32 MG (Fig. 4); at the strongest published field,

80 MG (near-perpendicular z - y), the model remains within tolerance. This is a substantially stronger statement than “qualitative agreement”: at these conditions the linear kinetic response, including its magnetized electron dynamics and eigenmode mixing, is quantitatively predictive across nearly two decades of field strength.

Two structural conclusions follow from the declared variants, evaluated at every condition (medians over carbon conditions at $B \geq 8$ MG). First, the single-broad-IAW reduction (unmagnetized ions) is indistinguishable from the full response *everywhere* on the carbon map (g -ratio 1.00–1.01 in all six geometry/polarization lanes; Fig. 1, dotted): at $\Delta\lambda = 0.9 \text{ \AA}$ the beat frequency sits far above the carbon ion-Bernstein band ($\Delta\omega/\Omega_{ci} \gtrsim 6$ even at 64 MG), so magnetized ion-mode structure plays no role at the IAW-resonant detuning. Second, the decomposition of the field response is cleanly geometric: at $\xi \approx 0^\circ$ and 45° the mode-mixing-only null model already reproduces the full prediction (g -ratio 1.00–1.01)—there the entire magnetization response, including the z - z suppression and z - y activation, is carried by the laser *eigenmode* structure (polarization mixing and Faraday dephasing of the coherent two-mode transfer)—whereas at $\xi \approx 88^\circ$ the null model overpredicts by a factor $\simeq 2.0$: there, and only there, the magnetized *electron* response of the beat wave supplies an additional suppression. This quantifies, condition by condition, the two regimes the original study identified qualitatively (laser-mode magnetization at near-parallel geometry versus plasma-wave magnetization at near-perpendicular geometry). The published eigenmode-resolved gains (X above O at large fields) are reproduced in ordering at 100 MG but not in the small splitting at 10 MG, where the measured splitting ($\lesssim 0.02$) is smaller than either model variant; eigenmode-level comparisons are accordingly treated as secondary diagnostics.

4.2 Near-perpendicular cross-polarized channel: the clean theory failure

The three marginal carbon conditions form one contiguous segment: $\xi \approx 88^\circ$, z - y , 10–32 MG, where the model predicts activated transfer $g \simeq 0.31$ –0.35 while the measurement stays at 0.02–0.12 (the two highest-field points, 50 and 80 MG, return within tolerance as the measured channel finally activates). The variant comparison is decisive about what the failure is *not*: the full magnetized-ion response and the single-broad-IAW reduction agree to better than 0.01 in this channel, so magnetized ion-mode emergence—the mechanism one might naturally suspect at near-perpendicular geometry—is excluded as the driver. The mode-mixing-only model overpredicts twice as strongly ($g \simeq 0.62$), so magnetized electron response supplies half of the needed suppression but not all of it. What remains is the coupling/drive structure itself: either the magnetization corrections to the ponderomotive drive that the theory drops (its own stated approximation, expected to degrade as ω_{ce}/ω_0 grows past ~ 0.03 –0.1 in exactly this range), or the electromagnetic part of the near-perpendicular beat response (fast magnetosonic channel, $\beta \sim 3 \rightarrow 0.1$ across this segment) that our electrostatic reconstruction omits—a gap we declared in advance. The data cannot distinguish these two, and we report the segment as category-(i)/reconstruction-gap with that explicit ambiguity rather than forcing a single label. Practically, the linear model’s *worst-case* error here is an overprediction of an absolutely small channel ($g \lesssim 0.35$): a conservative bound, not a misordering.

4.3 Hydrogen families and the temperature trend

The hydrogen conditions ($ZT_e/T_i = 4$: strongly Landau-damped IAW, oblique out-of-plane field, reduced numerical resolution, double intensity) are where the comparison gets genuinely harder, and the map says so (Fig. 4): 22 of 43 conditions hold, 20 are marginal, one breaks. The pattern is structured, not random. The z - z channel at $T_e = 0.4 \text{ keV}$ holds essentially everywhere (12/13) with the anchored model tracking the factor-two suppression over 0–30 MG; its z - y channel is overpredicted by 0.2–0.4 through the 7–28 MG activation region (with a single *breaks* point at 12.5 MG) before converging at the top of the scan. At $T_e = 1.2 \text{ keV}$ the model overpredicts the z - z channel by ~ 0.2 –0.5 *at all fields including* $B = 0$ —and a discrepancy already present without a field cannot be a magnetization failure. The temperature trend across families (anchored only once, at 0.4 keV) is overpredicted by $\sim 50\%$ at 1.2 keV. Candidate drivers here are comparison-level: these published scans used the reduced resolution (10 cells/ λ_0 , 5 particles/cell, $\Delta x/\lambda_{De} \simeq 4$ at 0.4 keV) and carry no time-spread error bars at 0.4 keV (single measurement time in the released files), and the strongly damped hydrogen quasimode makes the measured transfer sensitive to the evolving ion distribution. Our carbon temperature block and resolution validation (below) were designed to discriminate; the mechanism analysis (Sec. 5) folds in their outcome.

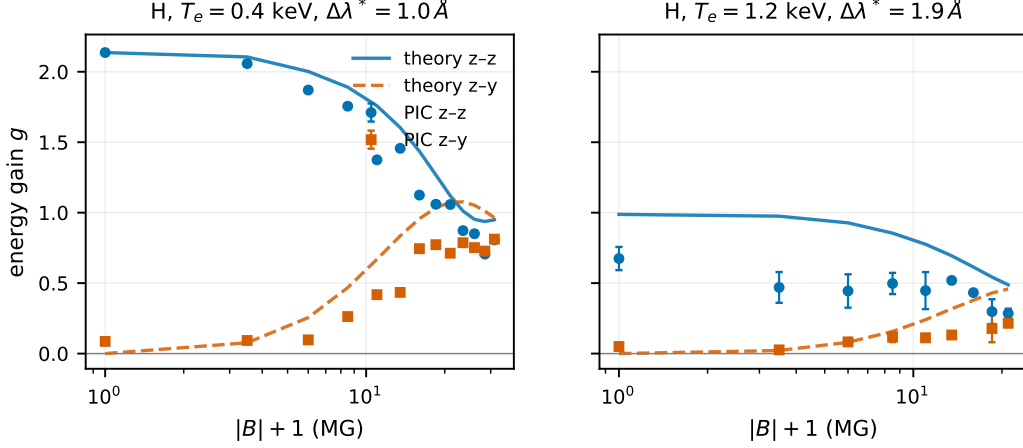


Figure 3: Hydrogen families (oblique out-of-plane field, $\theta_B = 45^\circ$, $\phi_B = 15^\circ$): measured gains at each family’s comparison detuning (symbols) against the anchored linear model (lines; single anchor at the 0.4 keV unmagnetized z - z condition shared by both temperatures). The z - z suppression at 0.4 keV tracks within tolerance; the cross-polarized activation band and the entire 1.2 keV z - z family are overpredicted (Sec. 5, segments B and C).

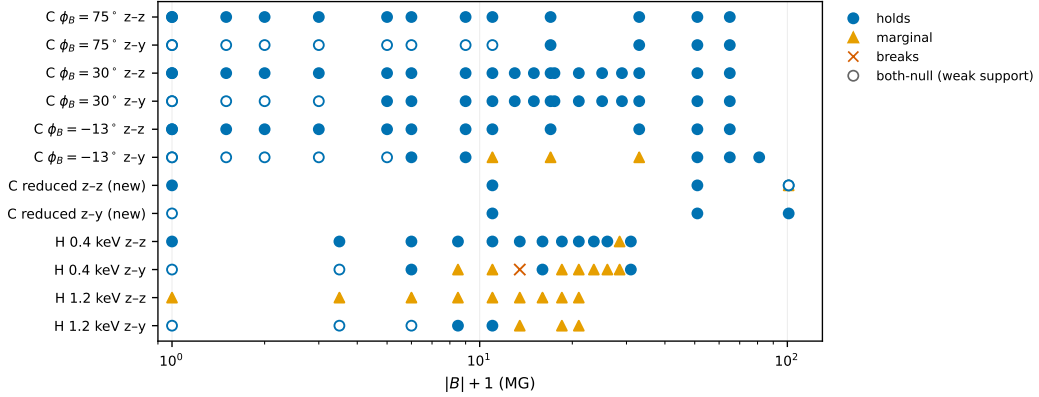


Figure 4: The validity map. Every analyzed condition, arranged by configuration lane (family, orientation, polarization) and field strength; symbol/color denote the classification under the pre-registered rule (Eq. 4); open symbols mark “both-null” agreements (weak evidence). New reduced-scale conditions (this work) occupy the C-reduced lanes and the intensity/temperature blocks are tabulated in Table 1 and the appendix.

4.4 Reduced-scale validation and the new conditions

What the reduced resolution can and cannot reproduce. The declared gate compared reduced-scale (10 cells/ λ_0 , 5 particles/cell) carbon runs against the full-resolution released values at $\phi_B = 30^\circ$, $B \in \{0, 10, 50\}$ MG. Outcome, reported in full: the *trend* criteria pass—the field ordering $g(0) > g(10 \text{ MG}) > g(50 \text{ MG})$ ($0.84 > 0.68 > 0.51$, seed-ensemble means) and the channel ordering $g_{z-y} < g_{z-z}$ at 10 MG ($0.28 < 0.68$) are reproduced, and the eigenmode ordering ($g_X = 0.63 > g_O = 0.29$ at 50 MG) matches the published full-resolution split—but the *absolute* criterion fails at $B = 0$: the reduced configuration recovers 0.84 ± 0.10 against the full-resolution 1.34 ± 0.11 (deficit 0.50, tolerance 0.30), while 10 and 50 MG pass (0.68 vs 0.95 ; 0.51 vs 0.72). The deficit fraction shrinks with field strength (recovery ratios 0.63, 0.72, 0.71), i.e., the coarse configuration under-recovers most where the transfer is largest. Two follow-ups localize the deficit. A diagnostic run with temperature output shows the electron temperature *drifts down* by only 8% over the measurement window (below the pre-registered escalation trigger), but the ion temperature climbs 39% between

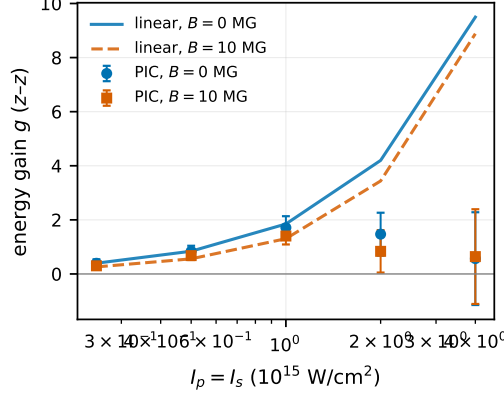


Figure 5: The intensity boundary: measured z - z gain (offset-corrected, symbols) against the anchored linear prediction (lines) for the joint pump-and-seed intensity scan at $B = 0$ and 10 MG. The linear description holds through $1 \times 10^{15} \text{ W/cm}^2$ and saturates far below the prediction beyond it; at the two highest drives the comparison’s own prerequisites fail as well (null-channel amplification, pump depletion, flux reversal; Sec. 5).

the baseline and final measurement ($169 \rightarrow 338 \text{ eV}$ from initialization), driven by particle noise at 5 particles/cell: ZT_e/T_i degrades from 24 to ≈ 10 during the run, softening the ion-acoustic response exactly where it is most resonance-sensitive (the specific ion-channel identification was made from this diagnostic after the validation residuals were known; the mechanism class—particle-noise fidelity, M7—was pre-registered). Consistently, the one-step-finer ladder configuration (10 particles/cell, halving the heating rate) restores both validation conditions to within the absolute tolerance: $0.84 \rightarrow 1.14$ at $B = 0$ (full resolution 1.34, difference 0.20) and $0.68 \rightarrow 0.70$ at 10 MG (0.95, difference 0.24). Consequently, and as pre-declared in the study log, all new-condition comparisons below use the reduced family’s own $B = 0$ anchor, so they test trends and relative responses at matched fidelity; their absolute-scale limitations are carried as an explicit reduced-fidelity (M7) caveat rather than hidden in widened error bars. Three-seed RNG ensembles give a seed spread (sample standard deviation) of 0.099 at $B = 0$ and 0.031 at 10 MG—below the pre-declared 0.3 escalation trigger—and the larger value is propagated to all new conditions. Under this protocol the six validation conditions themselves classify as *holds* (the $B = 0$ anchor trivially; 10 and 50 MG with $\delta/\tau = 0.37$ and 0.40; the cross-polarized channel at both fields within tolerance).

Intensity. The joint pump-and-seed intensity scan ($I = 0.25\text{--}4 \times 10^{15} \text{ W/cm}^2$ at $B \in \{0, 10\} \text{ MG}$, Fig. 5) brackets the linear window from both sides with a strikingly monotone progression. At 0.25×10^{15} the anchored linear prediction is exact to well within uncertainties ($\delta/\tau = 0.01$ and 0.15; inter-time drift consistent with zero)—the comparison is at its cleanest, with no sign of the noise-competition degradation that full-scale runs at halved pump intensity were reported to suffer. At 1×10^{15} (twice the reference drive) all three measured channels still hold ($\delta/\tau = 0.15\text{--}0.16$) although the inter-time drift grows to 0.47 at $B = 0$ —the earliest warning of steady-state loss. At 2×10^{15} the comparison itself begins to collapse: the null (z - y , $B = 0$) companion channel, which the background-offset protocol relies on, *amplifies* ($g_0 = +1.9 \pm 0.9$, against -0.05 and -0.01 at lower drive)—spontaneous side/backscatter has gone unstable, exactly the behavior the original study warned of when doubling the seed—while the pump-window energy drops 3–4% and the raw z - z gains saturate 19% below the linear scaling (3.39 vs 4.20 at $B = 0$); the lane classifies marginal ($\delta/\tau \simeq 1.7$) with the offset correction applied mechanically as pre-registered. At 4×10^{15} the collapse is total: the null channel gains $e^{3.6}$, the pump window is depleted 12–14%, and the domain-averaged Poynting flux *reverses sign* ($\langle S_x \rangle/I_s = -1.4$ at $B = 0$)—the simulation is backscatter-dominated and the lane breaks ($\delta/\tau = 2.1\text{--}2.6$). The practical boundary of the linear description on this axis therefore lies between 1 and $2 \times 10^{15} \text{ W/cm}^2$ at these plasma conditions; notably, by the time the theory’s *linearity* fails, the *measurement’s* own prerequisites (quiet null channel, undepleted pump, forward-flowing energy) have failed with it, so the boundary is jointly a theory and a comparison-validity limit (M5+M6, with the M8 drift signature preceding both).

Table 1: Representative conditions of the validity map (full table of all conditions in the appendix). g_{lin} : anchored linear kinetic prediction; g_{PIC} : measured gain (offset-corrected) with combined uncertainty; δ/τ : residual over tolerance (Eq. 4; ≤ 1 holds, ≥ 2 breaks); verdict superscript 0 marks both-null agreements. Mechanism labels are defined in Sec. 5.

	$ B $ (MG)	ϕ_B (deg)	n_e (10^{20} cm^{-3})	T_e (keV)	I (10^{15} W/cm^2)	pol.	g_{lin}	$g_{\text{PIC}} \pm \sigma$	δ/τ	verdict	mechanism
C	0	any	3.5	0.8	0.5	$z-z$	1.34	1.34 ± 0.15	0.00	holds	none
C	0	any	3.5	0.8	0.5	$z-y$	0.00	0.00 ± 0.09	0.02	holds ⁰	none
C	10	75	3.5	0.8	0.5	$z-z$	1.29	1.25 ± 0.15	0.13	holds	none
C	50	75	3.5	0.8	0.5	$z-z$	0.67	0.76 ± 0.15	0.28	holds	none
C	50	75	3.5	0.8	0.5	$z-y$	0.69	0.72 ± 0.10	0.15	holds	none
C	10	30	3.5	0.8	0.5	$z-z$	0.92	0.95 ± 0.14	0.11	holds	none
C	10	30	3.5	0.8	0.5	$z-y$	0.57	0.54 ± 0.09	0.15	holds	none
C	10	-13	3.5	0.8	0.5	$z-z$	0.43	0.36 ± 0.12	0.32	holds	none
C	10	-13	3.5	0.8	0.5	$z-y$	0.31	0.02 ± 0.10	1.53	marginal	M1/EM-channel
C	32	-13	3.5	0.8	0.5	$z-y$	0.32	0.12 ± 0.09	1.16	marginal	M1/EM-channel
H	0	15	3.0	0.4	1.0	$z-z$	2.14	2.14 ± 0.11	0.00	holds	none
H	10	15	3.0	0.4	1.0	$z-z$	1.76	1.37 ± 0.10	0.93	holds	none
H	12.5	15	3.0	0.4	1.0	$z-y$	0.83	0.43 ± 0.09	2.30	breaks	M1/EM-channel
H	0	15	3.0	1.2	1.0	$z-z$	0.99	0.67 ± 0.12	1.33	marginal	M7
H	10	15	3.0	1.2	1.0	$z-z$	0.78	0.45 ± 0.16	1.06	marginal	M7
C_r	0	30	3.5	0.8	0.5	$z-z$	0.84	0.84 ± 0.20	0.00	holds	none
C_r	10	30	3.5	0.8	0.5	$z-z$	0.56	0.68 ± 0.16	0.37	holds	none
C_r	50	30	3.5	0.8	0.5	$z-z$	0.39	0.51 ± 0.15	0.40	holds	none
C_r	10	30	3.5	0.8	1.0	$z-z$	1.30	1.40 ± 0.31	0.16	holds	none
C_r	10	30	3.5	0.8	2.0	$z-z$	3.44	0.84 ± 0.78	1.66	marginal	M5+M6
C_r	10	30	3.5	0.8	4.0	$z-z$	8.88	0.64 ± 1.75	2.35	breaks	M5+M6
C_r	0	30	3.5	0.8	4.0	$z-z$	9.50	0.57 ± 1.71	2.60	breaks	M5+M6
C_r	10	30	3.5	0.8	0.25	$z-z$	0.26	0.30 ± 0.13	0.15	holds	none
C_r	100	30	3.5	0.8	0.5	$z-z$	0.38	0.83 ± 0.17	1.31	marginal	M1
C_r	100	-13	3.5	0.8	0.5	$z-z$	0.18	0.18 ± 0.13	0.01	holds	none
C_r	10	30	3.5	0.4	0.5	$z-z$	0.95	0.77 ± 0.22	0.40	holds	none
C_r	10	30	3.5	1.2	0.5	$z-z$	0.32	0.50 ± 0.20	0.47	holds	none

Summary of all 165 cases: 134 hold, 27 marginal, 4 break under the pre-registered rule; dominant axes: I (NMI 0.32), beat-field angle (NMI 0.30). Full per-case table in the appendix.

100 MG. At $B = 100 \text{ MG}$ ($\omega_{ce}/\omega_0 = 0.33$) the theory's own drive window $\omega_{ce} \ll \omega_0$ is deliberately strained, and the map records its first *under*predictions. At the intermediate orientation the measured $z-z$ transfer exceeds the anchored linear prediction by a factor 2.2 (0.83 ± 0.17 vs 0.38; marginal, $\delta/\tau = 1.31$; the finer-fidelity confirmation run gives 0.64, a $1.7\times$ excess)—numerically consistent with the X-mode quiver enhancement $(1 - \omega_{ce}/\omega_0)^{-2} = 2.24$ that the averaged-frequency electrostatic-potential drive discards, and corroborated by the measured eigenmode split ($g_X = 1.08$ vs $g_O = 0.32$, the largest X/O asymmetry in the data set). The cross-polarized channel at the same orientation is also underpredicted but within tolerance (0.57 vs 0.38), and the near-perpendicular channel holds (0.18 vs 0.19), its response dominated by the magnetized-electron suppression. The near-parallel ($\phi_B = 75^\circ$) conditions—which would have provided the cleanest pre-registered test of the drive-window mechanism—could not be classified: at this field strength the two independent gain-extraction methods disagree grossly there (Fourier-window total 2.5 ± 4.9 versus eigenmode-reconstruction 0.74 ± 0.28 for $z-z$), a measurement-integrity anomaly we attribute to the strongly split eigenmode spectrum escaping the fixed Fourier windows; the two conditions are excluded from the map with that stated rationale rather than classified from either method (Sec. 6.3).

Carbon temperature block. The published data confound temperature with species, geometry, drive, and resolution; the new carbon block varies $T_e = 4T_i$ alone (0.4/0.8/1.2 keV at $B \in \{0, 10\} \text{ MG}$, matched fidelity, single anchor at 0.8 keV). *Every* condition holds. The measured $z-z$ trend at $B = 0$ is $1.13 \rightarrow 0.84 \rightarrow 0.67$ against predicted $1.40 \rightarrow 0.84 \rightarrow 0.48$: the model's temperature slope is somewhat too steep (25–28% over- and under-prediction at the cold and hot ends, $\delta/\tau = 0.4\text{--}0.7$), but the trend never leaves the tolerance band, at either field. Temperature alone, over this range and at matched fidelity, does *not* break the linear description—which sharpens the interpretation of the hydrogen segment (Sec. 5).

Table 2: Single-axis separability of the validity classification (holds vs. not): best one-dimensional threshold accuracy and normalized mutual information (NMI) per axis.

axis	cases	best 1-D accuracy	threshold	NMI
I	165	0.84	1	0.32
$ \cos \xi $ (beat-field angle)	139	0.80	0.0142	0.30
T_e	165	0.83	0.8	0.17
$ B $	165	0.81	0 MG	0.13
ω_{ce}/ω_0	165	0.81	0	0.13
polarization	165	0.81	0	0.00
base rate (majority class) = 0.81				

4.5 Compactness: which variables carry the signal

Is the validity boundary a tangled surface in the full parameter space, or do a few coordinates carry it? Table 2 quantifies single-axis separability of the classification over all analyzed conditions: for each axis, the accuracy of the best one-dimensional threshold and the normalized mutual information (NMI) with the holds/not-holds label. Two coordinates dominate. *Drive intensity* carries the most signal (NMI 0.32), and its optimal single threshold falls exactly at $1 \times 10^{15} \text{ W/cm}^2$ —the measured linearity/comparison boundary; part of this signal, we note, is the aliasing of the hydrogen families (at 1×10^{15}) with their geometry-driven marginals. The *beat-field angle* is second (NMI 0.30): failures cluster at $|\cos \xi| \lesssim 0.3$, i.e., oblique-to-perpendicular beat geometry, which is where the cross-polarized-coupling overprediction lives. Electron temperature retains a modest residual signal (NMI 0.17) that the matched-fidelity carbon block shows is not causal at these conditions (Sec. 4.4); polarization channel carries essentially none (NMI 0.001)—the model is equally good or bad for both channels once geometry and drive are fixed, a useful negative result. Most strikingly, *field strength itself* is nearly inert (NMI 0.13, no useful single threshold): between 0.5 and 80 MG the linear description does not degrade with $|B|$ at fixed geometry, and the only field-driven boundary appears when ω_{ce}/ω_0 approaches the drive window’s edge at 100 MG. The practical summary is a two-coordinate rule—check the drive strength against the saturation threshold and the beat-field angle against $|\cos \xi| \sim 0.3$ —rather than a high-dimensional surface; the map is, in this sense, compact.

5 Mechanism analysis

Attribution follows the pre-registered menu and diagnostics: candidate failures of the theory’s physical assumptions—(i) the averaged-frequency fluid drive without magnetization corrections, valid only in the window $\Delta\omega \ll \omega_{ce} \ll \omega_0$; (ii) the linear, ponderomotive-free ion response; (iii) mediation by a single strongly damped acoustic resonance computed along unperturbed orbits—versus failures of comparison prerequisites: steady state (M8/kernel), undepleted pump (M5), negligible secondary scatter (M6), adequate numerical resolution (M7). Contiguous non-holding conditions within a configuration lane form a segment; each segment receives the single best-supported label, or an explicit ambiguity.

Segment A: carbon, $\xi \approx 88^\circ$, z - y , 10–32 MG (3 marginal conditions). The evidence chain of Sec. 4.2: variant agreement $|g_{\text{full}} - g_{\text{IAW-only}}| < 0.01$ excludes magnetized-ion-mode emergence (the assumption-(iii) mechanism one would first suspect at this geometry); the mode-mixing-only null model doubles the overprediction, so magnetized electron response is necessary but insufficient; the measured channel activates only at 50–80 MG. Attribution: assumption-(i) drive/coupling corrections at $\omega_{ce}/\omega_0 = 0.03$ –0.11 or the electromagnetic near-perpendicular beat channel absent from our electrostatic reconstruction (a declared reconstruction gap; the plasma crosses from $\beta \sim 3$ to ~ 0.3 across the segment)—reported as unresolved between these two. The specific decision branch combining these pre-stated diagnostics was formalized after inspecting the variant comparison and is disclosed as such in the analysis log.

Segment B: hydrogen 0.4 keV, z - y , 7.5–27.5 MG (8 conditions, one break). Same signature as segment A—overprediction of the activated cross-polarized channel, $V_{\text{full}} \simeq V_{\text{IAW-only}}$ —at an oblique out-of-plane geometry ($\xi \approx 72^\circ$), reinforcing that the failure follows the cross-polarized

coupling, not the ion response. Reduced numerical resolution (M7) was the pre-registered alternative, and the carbon reduced-scale runs discriminate against it: the resolution deficit on *magnetized* conditions is bounded at 26–28% by the validation set (rising to 37% only at $B = 0$), and, more tellingly, the reduced-scale cross-polarized activation at 10 MG, $\xi = 45^\circ$ ($g = 0.28 \pm 0.15$) agrees with its matched-fidelity anchored prediction (0.33 , $\delta/\tau = 0.19$)—at moderate beat–field angle the activated channel shows no anomaly even at reduced fidelity. The overprediction therefore tracks the beat–field *angle* (present at $\xi \approx 72^\circ$ – 88° , absent at 45°), consistent with the segment-A coupling mechanism rather than with resolution.

Segment C: hydrogen 1.2 keV, z – z , all fields including $B = 0$ (9 conditions). A discrepancy present at $B = 0$ cannot be a magnetization-physics failure; this segment is a failure of the *temperature trend* under the shared-anchor protocol (the model, anchored at 0.4 keV, overpredicts the 1.2 keV family by $\sim 50\%$). Pre-registered candidates: comparison-level artifacts of the reduced resolution and single-time measurement in these released files (M7), and evolving-distribution effects in the strongly damped hydrogen quasimode (M8/(iii)). The carbon temperature block was designed as the discriminator, and it discriminates: at matched species, geometry, drive, and fidelity, the same protocol tracks the carbon $0.4 \rightarrow 1.2$ keV trend within tolerance everywhere (with a mild slope error of *opposite sign* at the hot end: the model underpredicts hot carbon but overpredicts hot hydrogen). Temperature per se is therefore not the failing ingredient; what distinguishes hydrogen is its strongly Landau-damped quasimode ($ZT_e/T_i = 4$ against carbon’s 24), whose response is damping-dominated and hence first-order sensitive to the evolving ion distribution (PIC ion heating flattens the resonance during the measurement—directly observed in our temperature diagnostic, where T_i rose 39% over the window). We attribute the segment to M8/(iii) evolving-distribution physics acting on a strongly damped quasimode, with reduced-fidelity ion noise (M7) as an inseparable amplifier of the same effect—a combined attribution rather than a clean single label, and we say so.

Segment D: intensity lanes, $I \geq 2 \times 10^{15}$ W/cm² (6 conditions: 3 marginal, 3 breaks). The pre-registered discrimination among M4 (ion ponderomotive nonlinearity), M5 (depletion), M6 (secondary instabilities), and M8 (steady-state loss) is resolved by the run diagnostics in a definite order. M8 arrives first: the inter-time gain drift grows monotonically with drive (≈ 0 at 0.25; 0.13 at 0.5; 0.47 at 1; up to 1.5 at 2×10^{15}) while the gains still hold. M6 fires between 1 and 2×10^{15} : the null-channel companion gain jumps from -0.01 to $+1.9$ and then $+3.6$, so spontaneous scattering competes with, then dominates, the seeded transfer. M5 becomes measurable at the same threshold (pump-window energy down 3–4%, then 12–14%) and the domain-averaged energy flux reverses at 4×10^{15} . A clean attribution of the *remaining* sub-linear saturation to the theory-side ion nonlinearity (M4) is therefore not possible from these runs—by the time the linear prediction fails, the comparison’s own prerequisites have failed with it, and the lane is attributed to the joint M5+M6 collapse (category: comparison validity, with concurrent theory-linearity loss). This is itself the practically relevant statement: above $\sim 1 \times 10^{15}$ W/cm² at these conditions neither the linear theory *nor* this measurement protocol describes the interaction.

Segment E: 100 MG, intermediate orientation (1 marginal). The drive-window probe (M1, pre-registered) at $\omega_{ce}/\omega_0 = 0.33$: the measured z – z transfer exceeds the anchored prediction by $2.2\times$ at the survey fidelity and $1.7\times$ in the finer-fidelity confirmation run—robust in sign and scale, and bracketing the X-mode quiver enhancement $(1 - \omega_{ce}/\omega_0)^{-2} = 2.24$ dropped by the theory’s averaged-frequency drive; the largest measured X/O splits of the data set ($g_X/g_O \simeq 3$ at both fidelities) corroborate an X-channel origin. A short detuning scan at this field shows the response already declining by $\Delta\lambda = 1.5$ Å and negative at 2.1 Å, so the excess at 0.9 Å is not an artifact of a strongly shifted resonance peak. The pre-registered cleanest version of this test—residual tracking ω_{ce}/ω_0 at the near-parallel orientation—could not be evaluated because both near-parallel conditions at this field are measurement-anomalous (Sec. 6.3); the attribution therefore rests on the intermediate-orientation magnitude bracket and is labeled M1-consistent rather than M1-confirmed.

Non-failures worth recording. Under assumption (iii) one expects additional narrow resonances near perpendicular geometry; within the carbon z – z data the single-broad-IAW reduction never separates from the full response by more than the uncertainties, so *at the IAW-resonant detuning and $B \leq 80$ MG the single-mode picture is not the operative approximation error*—consistent with the beat frequency lying far above the carbon ion-Bernstein band at these fields. The finite-duration correction, by contrast, is *not* a cosmetic rescaling: re-evaluating the map with the ideal steady-state

response (same anchoring protocol) flips the classification of 107 of 136 conditions, with a median anchored-gain shift of 0.57—because a $B = 0$ anchor taken on the narrow ideal resonance forces an enormous drive scale that then overshoots every broad-response condition. At these damping rates ($\nu T \ll 1$ for the carbon IAW) a picosecond-scale measurement simply does not sample the steady-state response, and any quantitative use of linear CBET theory against short-pulse kinetic simulation or experiment must model the drive duration; this comparison-validity requirement (the M8/steady-state prerequisite of Sec. 5) is one of the sharpest methodological findings of the map. The residual detuning skew the kernel leaves unexplained is discussed below.

6 Discussion

6.1 Implications for practice

For ICF-relevant modeling the map reads as a usage rule. In the regime that dominates applications—moderately magnetized ($\lesssim$ tens of MG), IAW-mediated transfer between like-polarized beams at drive intensities within the linear window—the linearized Vlasov description, evaluated with full magnetized susceptibilities and eigenmode geometry and calibrated at one unmagnetized point, is quantitatively predictive (30% class) across field strength and orientation; design tools can use it in place of kinetic simulation. Care is needed in three places. First, the activated cross-polarized channel near perpendicular beat geometry is overpredicted by up to a factor of several; since this channel is absolutely small there ($g \lesssim 0.35$ at these conditions), linear estimates provide a conservative upper bound rather than a reliable value. Second, predictions become unreliable once the drive exceeds $\sim 1 \times 10^{15} \text{ W/cm}^2$ at these conditions ($I_{15} \lambda_{\mu m}^2 \simeq 0.12$), where the measured transfer saturates below the linear scaling and spontaneous scattering competes; existing unmagnetized saturation criteria [9–11, 13] appear to carry over as the practical guardrail. Third, the temperature dependence across plasma families is the softest part of the linear prediction in the present evidence; until the trend test is settled with matched-resolution kinetic data, temperature extrapolations should carry an uncertainty of order the $\sim 50\%$ discrepancy we measure. Where fields reach $\omega_{ce}/\omega_0 \gtrsim 0.2$ – 0.3 ($\gtrsim 60 \text{ MG}$ at 3ω), the averaged-frequency drive itself becomes the limiting ingredient and errs on the *conservative* side—the measured transfer exceeds the linear prediction by the dropped quiver-enhancement factor, up to $\sim 2\times$ at 100 MG —so linear estimates there are lower bounds, and polarization-resolved modeling (the enhancement is X-mode-specific) is the needed refinement.

6.2 Relation to prior work

Our anchored-comparison protocol operationalizes, for the magnetized problem, the accuracy questions long studied for unmagnetized CBET: the polarization anisotropy and saturation experiments of Turnbull *et al.* [9], the kinetic/nonlinear corrections of Seaton *et al.* [11] and Yin *et al.* [12], ion-heating saturation [10], and the transient sub-ns response of Xu *et al.* [13], whose trapping-induced red-shift of maximum transfer is a natural candidate for the residual detuning skew we observe. On the magnetized side, the map complements the MagCBET theory [28] and its qualitative PIC validation [29] by quantifying where those predictions are numerically trustworthy, and connects to broader magnetized-LPI results—kinetic magnetized scattering theory [32–35], magnetized SRS mitigation [24], strongly magnetized coupling and absorption [25, 26], three-wave benchmarking in magnetized kinetic simulation [31], and the 2026 magnetized multi-beam LPI study of Lacoste *et al.* [27]. The validity-map framing itself—pre-registered tolerances, per-condition support, mechanism attribution against a declared menu—is, to our knowledge, new in this context and is directly reusable for the non-IAW magnetized channels (Bernstein/lower-hybrid mediation) that Ref. [29] announces as future work.

6.3 Failure analysis and limitations

Honesty about what did not work sharpens the map’s edges. (1) The reconstruction underpredicts absolute gains by $2.7\times$ (carbon) and $1.7\times$ (hydrogen); the finite-duration response model we built is evidently only $O(1)$ accurate, and a first-principles space–time transient calculation (or matched-duration PIC at several drive lengths) is the obvious next step. Because a single anchor absorbs it, none of the mapped *trends* depend on this factor, but absolute-scale users should know it is there. (2) The measured unmagnetized gain curve is skewed ~ 0.2 – $0.3 \text{ \AA}$ below the modeled resonance; candidate

explanations (trapping red-shift [13], seed angular spectrum, evolving ion temperature) remain untested here. (3) Eigenmode-resolved (X/O) gain splits at intermediate fields are not reproduced in detail by the coherent two-mode model, and we used them only as secondary evidence; the frozen-pump approximation and the eigenmode-energy bookkeeping of the measurement itself deserve a dedicated study. (4) The hydrogen temperature-trend segment received a combined attribution (evolving-distribution physics on a strongly damped quasimode, amplified by reduced-fidelity ion noise) rather than a single label; separating the two cleanly would need matched-fidelity hydrogen reruns at several particle counts, which the run budget did not cover. (5) Statistical support is thin exactly where the physics is most interesting: the near-perpendicular z - y channel involves small gains near the background-offset systematic, and the 0.4 keV hydrogen files carry no time-spread error bars; both are flagged in the per-condition table rather than smoothed over. (6) Two measurement-machinery failures surfaced at the edges of the map and are excluded or flagged rather than classified: at 100 MG, $\phi_B = 75^\circ$, the fixed-window Fourier gain and the eigenmode-reconstruction gain disagree grossly (spread up to 4.9), consistent with the strongly split eigenmode spectrum escaping the windows tuned for weak splitting; and at $I \geq 2 \times 10^{15}$ W/cm² the background-offset protocol itself collapses because its null channel amplifies—the offset is then subtracted mechanically as pre-registered, but those verdicts measure the collapse of the comparison as much as of the theory. (7) Axes we considered and dropped (density, crossing angle, a third polarization state) are recorded in the study log; polarization carries almost no validity signal within the two channels tested (Table 2), which is itself useful negative information.

6.4 Computational cost

The complete new-simulation campaign comprised 49 EPOCH 2D-3V runs (10 validation/ensemble, 15 intensity, 6 at 100 MG, 8 temperature, 3 detuning/ seed stretch, 6 finer-fidelity ladder/confirmation, 1 heating diagnostic): 26.1 aggregate wall-hours on 16-core groups, 418 core-hours in total, at ~ 8.5 core-hours per reduced-scale condition (~ 29 minutes on 16 cores; the paper’s full-resolution configuration costs $16\times$ more per run). The reconstructed linear baseline evaluated all 172 map conditions $\times$ 7 model variants in under three core-hours—roughly 2 CPU-seconds per condition-variant including the full Bessel-sum susceptibilities—a per-condition asymmetry of order 10^4 . That asymmetry is the practical argument for this map: five more PIC runs would extend it only marginally (the highest-value additions we did not make are matched-fidelity hydrogen particle-count scans to split the segment-C attribution, and a $\phi_B = 75^\circ$ re-measurement at 100 MG with windows adapted to the split eigenmode spectrum), whereas the linear model can now be applied across its mapped domain at negligible cost.

6.5 Conclusions

Linearized Vlasov modeling of magnetized CBET is quantitatively trustworthy over a usefully large domain: at reference ICF underdense conditions it predicts the measured transfer within 30% across two decades of field strength, all orientations tested, and both polarization channels, once a single unmagnetized calibration point and the finite-duration measurement physics are accounted for. Its failures are structured and physical: the cross-polarized coupling near perpendicular beat geometry (a drive-structure or electromagnetic-channel effect, *not* ion-mode emergence), the intensity boundary of linearity, and an unresolved temperature-trend discrepancy inherited from mixed-fidelity validation data. A one-line rule for practitioners: *check the beat-field angle and the drive strength first; field strength itself is mostly harmless until $\omega_{ce}/\omega_0 \sim 0.1$.*

Reproducibility

All analysis code, EPOCH input decks, run logs, processed outputs, and the pre-registered protocol (with commit hashes) accompany this manuscript.

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A Verification of the reconstructed baseline and the analysis port

Susceptibilities. The magnetized susceptibility implementation (Eq. 2) satisfies $\sum_n e^{-b} I_n(b) = 1$ to machine precision for b up to 2.3×10^4 (the largest value occurring in the map, at 0.5 MG near-perpendicular), and converges to the unmagnetized Z -function form to better than 10^{-7} relative in the weak-field limit at oblique angles. The $k_{\parallel} \rightarrow 0$ limit reproduces the real Bernstein form.

Eigenmodes. Two independent implementations were used: a cold Stix solver (baseline) and a line-by-line port of the warm-fluid solver shipped with the original analysis code [30, 39] (used by the gain-extraction pipeline for exact convention parity). At all tested conditions (2–100 MG, all orientations) the two agree in refractive indices to $< 10^{-3}$ and in polarization vectors to overlap $|\langle e_{\text{warm}} | e_{\text{cold}} \rangle|^2 = 1.000$, confirming branch identification (branch 1 = X/R-like, smaller n^2).

Two-mode transfer. The coherent propagator (Eq. 3) reproduces the exact $B = 0$ nulls (zero cross-polarized transfer; equal circular gains) and reduces to independent-mode $\ln[(e^G - 1)/G]$ energy gains in the dephased limit. The scalar drive prefactor matches the standard unmagnetized kinetic gain formula [2, 9] to within the stated $\lesssim 8\%$ geometry/convention factor.

Analysis pipeline port. The Python port of the released MATLAB gain-extraction pipeline passes: the geometry-transform unit tests shipped with the original code; window/filter and shift-interpolation tests; and an end-to-end synthetic test in which two eigenmode fields with prescribed gain exponents (G_1, G_2) on the production grid are recovered to < 0.05 absolute in both branch energy gains by the full filter–integrate–shift–decompose chain. Fourier-integral and real-space eigenmode totals agree on raw gains to a median $|\Delta g| = 0.012$ across the released carbon scans, matching their agreement in the released files themselves.

B Full per-condition table

	$ B $ (MG)	ϕ_B (deg)	n_e	T_e (keV)	I	pol.	g_{lin}	$g_{\text{PIC}} \pm \sigma$	δ/τ	verdict	mech.
C	0	any	3.5	0.8	0.5	$z-y$	0.00	0.00 ± 0.09	0.02	holds ⁰	none
C	0	any	3.5	0.8	0.5	$z-z$	1.34	1.34 ± 0.15	0.00	holds	none
C	0	-13	3.5	0.8	0.5	$z-y$	0.00	0.00 ± 0.09	0.02	holds ⁰	none
C	0.5	-13	3.5	0.8	0.5	$z-y$	0.00	-0.01 ± 0.09	0.09	holds ⁰	none
C	1	-13	3.5	0.8	0.5	$z-y$	0.01	0.01 ± 0.09	0.03	holds ⁰	none
C	2	-13	3.5	0.8	0.5	$z-y$	0.05	0.04 ± 0.09	0.07	holds ⁰	none
C	4	-13	3.5	0.8	0.5	$z-y$	0.14	0.11 ± 0.08	0.19	holds ⁰	none
C	5	-13	3.5	0.8	0.5	$z-y$	0.18	0.15 ± 0.08	0.20	holds	none
C	8	-13	3.5	0.8	0.5	$z-y$	0.26	0.09 ± 0.09	0.90	holds	none
C	10	-13	3.5	0.8	0.5	$z-y$	0.31	0.02 ± 0.10	1.53	marginal	M1/EM-channel
C	16	-13	3.5	0.8	0.5	$z-y$	0.35	0.08 ± 0.09	1.47	marginal	M1/EM-channel
C	32	-13	3.5	0.8	0.5	$z-y$	0.32	0.12 ± 0.09	1.16	marginal	M1/EM-channel
C	50	-13	3.5	0.8	0.5	$z-y$	0.31	0.14 ± 0.10	0.81	holds	none
C	64	-13	3.5	0.8	0.5	$z-y$	0.30	0.16 ± 0.10	0.71	holds	none
C	80	-13	3.5	0.8	0.5	$z-y$	0.30	0.17 ± 0.08	0.76	holds	none
C	0	-13	3.5	0.8	0.5	$z-z$	1.34	1.34 ± 0.15	0.00	holds	none
C	0.5	-13	3.5	0.8	0.5	$z-z$	1.33	1.27 ± 0.16	0.16	holds	none
C	1	-13	3.5	0.8	0.5	$z-z$	1.29	1.31 ± 0.18	0.05	holds	none
C	2	-13	3.5	0.8	0.5	$z-z$	1.19	1.25 ± 0.14	0.15	holds	none
C	4	-13	3.5	0.8	0.5	$z-z$	0.96	1.04 ± 0.12	0.27	holds	none
C	5	-13	3.5	0.8	0.5	$z-z$	0.82	0.95 ± 0.11	0.46	holds	none
C	8	-13	3.5	0.8	0.5	$z-z$	0.53	0.55 ± 0.09	0.08	holds	none
C	10	-13	3.5	0.8	0.5	$z-z$	0.43	0.36 ± 0.12	0.32	holds	none
C	16	-13	3.5	0.8	0.5	$z-z$	0.33	0.26 ± 0.09	0.38	holds	none
C	32	-13	3.5	0.8	0.5	$z-z$	0.31	0.19 ± 0.10	0.67	holds	none
C	50	-13	3.5	0.8	0.5	$z-z$	0.31	0.19 ± 0.11	0.53	holds	none
C	64	-13	3.5	0.8	0.5	$z-z$	0.31	0.17 ± 0.10	0.70	holds	none
C	0	30	3.5	0.8	0.5	$z-y$	0.00	0.00 ± 0.09	0.02	holds ⁰	none
C	0.5	30	3.5	0.8	0.5	$z-y$	0.00	-0.01 ± 0.09	0.06	holds ⁰	none
C	1	30	3.5	0.8	0.5	$z-y$	0.01	-0.00 ± 0.09	0.09	holds ⁰	none
C	2	30	3.5	0.8	0.5	$z-y$	0.05	0.05 ± 0.09	0.01	holds ⁰	none
C	4	30	3.5	0.8	0.5	$z-y$	0.17	0.12 ± 0.09	0.29	holds	none
C	5	30	3.5	0.8	0.5	$z-y$	0.24	0.21 ± 0.08	0.21	holds	none
C	8	30	3.5	0.8	0.5	$z-y$	0.46	0.38 ± 0.09	0.42	holds	none
C	10	30	3.5	0.8	0.5	$z-y$	0.57	0.54 ± 0.09	0.15	holds	none
C	12	30	3.5	0.8	0.5	$z-y$	0.64	0.76 ± 0.15	0.37	holds	none
C	14	30	3.5	0.8	0.5	$z-y$	0.68	0.81 ± 0.15	0.42	holds	none
C	16	30	3.5	0.8	0.5	$z-y$	0.69	0.66 ± 0.10	0.13	holds	none
C	16.5	30	3.5	0.8	0.5	$z-y$	0.68	0.83 ± 0.15	0.51	holds	none
C	20	30	3.5	0.8	0.5	$z-y$	0.64	0.84 ± 0.15	0.65	holds	none
C	24	30	3.5	0.8	0.5	$z-y$	0.60	0.81 ± 0.14	0.72	holds	none
C	28	30	3.5	0.8	0.5	$z-y$	0.61	0.83 ± 0.14	0.77	holds	none
C	32	30	3.5	0.8	0.5	$z-y$	0.63	0.67 ± 0.09	0.21	holds	none
C	50	30	3.5	0.8	0.5	$z-y$	0.62	0.69 ± 0.09	0.35	holds	none
C	64	30	3.5	0.8	0.5	$z-y$	0.61	0.76 ± 0.09	0.64	holds	none
C	0	30	3.5	0.8	0.5	$z-z$	1.34	1.34 ± 0.15	0.00	holds	none
C	0.5	30	3.5	0.8	0.5	$z-z$	1.34	1.25 ± 0.15	0.23	holds	none
C	1	30	3.5	0.8	0.5	$z-z$	1.33	1.26 ± 0.16	0.20	holds	none
C	2	30	3.5	0.8	0.5	$z-z$	1.32	1.26 ± 0.15	0.16	holds	none
C	4	30	3.5	0.8	0.5	$z-z$	1.26	1.23 ± 0.15	0.08	holds	none
C	5	30	3.5	0.8	0.5	$z-z$	1.22	1.17 ± 0.15	0.14	holds	none
C	8	30	3.5	0.8	0.5	$z-z$	1.05	1.08 ± 0.15	0.12	holds	none
C	10	30	3.5	0.8	0.5	$z-z$	0.92	0.95 ± 0.14	0.11	holds	none
C	12	30	3.5	0.8	0.5	$z-z$	0.79	1.05 ± 0.19	0.68	holds	none
C	14	30	3.5	0.8	0.5	$z-z$	0.70	0.92 ± 0.18	0.62	holds	none
C	16	30	3.5	0.8	0.5	$z-z$	0.64	0.67 ± 0.09	0.12	holds	none
C	16.5	30	3.5	0.8	0.5	$z-z$	0.64	0.83 ± 0.14	0.67	holds	none
C	20	30	3.5	0.8	0.5	$z-z$	0.63	0.83 ± 0.14	0.70	holds	none
C	24	30	3.5	0.8	0.5	$z-z$	0.66	0.85 ± 0.15	0.65	holds	none
C	28	30	3.5	0.8	0.5	$z-z$	0.65	0.86 ± 0.15	0.71	holds	none
C	32	30	3.5	0.8	0.5	$z-z$	0.63	0.68 ± 0.10	0.25	holds	none
C	50	30	3.5	0.8	0.5	$z-z$	0.61	0.72 ± 0.10	0.48	holds	none

	$ B $ (MG)	ϕ_B (deg)	n_e	T_e (keV)	I	pol.	g_{lin}	$g_{PIC} \pm \sigma$	δ/τ	verdict	mech.
C	64	30	3.5	0.8	0.5	$z-z$	0.61	0.78 ± 0.09	0.74	holds	none
C	0	75	3.5	0.8	0.5	$z-y$	0.00	0.00 ± 0.09	0.02	holds ⁰	none
C	0.5	75	3.5	0.8	0.5	$z-y$	0.00	-0.01 ± 0.09	0.05	holds ⁰	none
C	1	75	3.5	0.8	0.5	$z-y$	0.00	-0.02 ± 0.09	0.09	holds ⁰	none
C	2	75	3.5	0.8	0.5	$z-y$	0.00	-0.02 ± 0.09	0.12	holds ⁰	none
C	4	75	3.5	0.8	0.5	$z-y$	0.02	-0.00 ± 0.09	0.11	holds ⁰	none
C	5	75	3.5	0.8	0.5	$z-y$	0.03	-0.01 ± 0.09	0.19	holds ⁰	none
C	8	75	3.5	0.8	0.5	$z-y$	0.07	0.04 ± 0.09	0.12	holds ⁰	none
C	10	75	3.5	0.8	0.5	$z-y$	0.10	0.09 ± 0.09	0.04	holds ⁰	none
C	16	75	3.5	0.8	0.5	$z-y$	0.23	0.21 ± 0.09	0.09	holds	none
C	32	75	3.5	0.8	0.5	$z-y$	0.56	0.56 ± 0.09	0.01	holds	none
C	50	75	3.5	0.8	0.5	$z-y$	0.69	0.72 ± 0.10	0.15	holds	none
C	64	75	3.5	0.8	0.5	$z-y$	0.61	0.79 ± 0.10	0.77	holds	none
C	0	75	3.5	0.8	0.5	$z-z$	1.34	1.34 ± 0.15	0.00	holds	none
C	0.5	75	3.5	0.8	0.5	$z-z$	1.34	1.26 ± 0.16	0.21	holds	none
C	1	75	3.5	0.8	0.5	$z-z$	1.34	1.27 ± 0.14	0.19	holds	none
C	2	75	3.5	0.8	0.5	$z-z$	1.34	1.29 ± 0.15	0.12	holds	none
C	4	75	3.5	0.8	0.5	$z-z$	1.33	1.25 ± 0.16	0.22	holds	none
C	5	75	3.5	0.8	0.5	$z-z$	1.33	1.25 ± 0.15	0.21	holds	none
C	8	75	3.5	0.8	0.5	$z-z$	1.31	1.27 ± 0.16	0.11	holds	none
C	10	75	3.5	0.8	0.5	$z-z$	1.29	1.25 ± 0.15	0.13	holds	none
C	16	75	3.5	0.8	0.5	$z-z$	1.23	1.22 ± 0.15	0.02	holds	none
C	32	75	3.5	0.8	0.5	$z-z$	0.93	0.94 ± 0.12	0.00	holds	none
C	50	75	3.5	0.8	0.5	$z-z$	0.67	0.76 ± 0.15	0.28	holds	none
C	64	75	3.5	0.8	0.5	$z-z$	0.71	0.89 ± 0.14	0.68	holds	none
C _r	100	-13	3.5	0.8	0.5	$z-y$	0.18	0.14 ± 0.13	0.16	holds	none
C _r	100	-13	3.5	0.8	0.5	$z-z$	0.18	0.18 ± 0.13	0.01	holds	none
C _r	0	30	3.5	0.8	0.25	$z-y$	-0.00	0.00 ± 0.13	0.00	holds ⁰	none
C _r	0	30	3.5	0.4	0.5	$z-y$	-0.00	0.00 ± 0.13	0.00	holds ⁰	none
C _r	0	30	3.5	0.8	0.5	$z-y$	-0.00	0.00 ± 0.13	0.00	holds ⁰	none
C _r	0	30	3.5	1.2	0.5	$z-y$	-0.00	0.00 ± 0.13	0.00	holds ⁰	none
C _r	0	30	3.5	0.8	1.0	$z-y$	0.00	0.00 ± 0.13	0.00	holds ⁰	none
C _r	0	30	3.5	0.8	2.0	$z-y$	0.00	0.00 ± 0.49	0.00	holds ⁰	none
C _r	0	30	3.5	0.8	4.0	$z-y$	0.00	0.00 ± 0.33	0.00	holds ⁰	none
C _r	10	30	3.5	0.4	0.5	$z-y$	0.61	0.48 ± 0.18	0.35	holds	none
C _r	10	30	3.5	0.8	0.5	$z-y$	0.33	0.28 ± 0.15	0.19	holds	none
C _r	10	30	3.5	1.2	0.5	$z-y$	0.18	0.24 ± 0.13	0.21	holds	none
C _r	10	30	3.5	0.8	1.0	$z-y$	0.83	0.76 ± 0.23	0.15	holds	none
C _r	10	30	3.5	0.8	2.0	$z-y$	2.23	0.17 ± 0.56	1.84	marginal	M5+M6
C _r	10	30	3.5	0.8	4.0	$z-y$	6.20	0.30 ± 1.42	2.08	breaks	M5+M6
C _r	50	30	3.5	0.8	0.5	$z-y$	0.39	0.32 ± 0.15	0.24	holds	none
C _r	100	30	3.5	0.8	0.5	$z-y$	0.38	0.57 ± 0.15	0.60	holds	none
C _r	0	30	3.5	0.8	0.25	$z-z$	0.40	0.40 ± 0.15	0.01	holds	none
C _r	0	30	3.5	0.4	0.5	$z-z$	1.40	1.13 ± 0.29	0.48	holds	none
C _r	0	30	3.5	0.8	0.5	$z-z$	0.84	0.84 ± 0.20	0.00	holds	none
C _r	0	30	3.5	1.2	0.5	$z-z$	0.48	0.67 ± 0.15	0.62	holds	none
C _r	0	30	3.5	0.8	1.0	$z-z$	1.85	1.71 ± 0.42	0.16	holds	none
C _r	0	30	3.5	0.8	2.0	$z-z$	4.20	1.47 ± 0.79	1.72	marginal	M5+M6
C _r	0	30	3.5	0.8	4.0	$z-z$	9.50	0.57 ± 1.71	2.60	breaks	M5+M6
C _r	10	30	3.5	0.8	0.25	$z-z$	0.26	0.30 ± 0.13	0.15	holds	none
C _r	10	30	3.5	0.4	0.5	$z-z$	0.95	0.77 ± 0.22	0.40	holds	none
C _r	10	30	3.5	0.8	0.5	$z-z$	0.56	0.68 ± 0.16	0.37	holds	none
C _r	10	30	3.5	1.2	0.5	$z-z$	0.32	0.50 ± 0.20	0.47	holds	none
C _r	10	30	3.5	0.8	1.0	$z-z$	1.30	1.40 ± 0.31	0.16	holds	none
C _r	10	30	3.5	0.8	2.0	$z-z$	3.44	0.84 ± 0.78	1.66	marginal	M5+M6
C _r	10	30	3.5	0.8	4.0	$z-z$	8.88	0.64 ± 1.75	2.35	breaks	M5+M6
C _r	50	30	3.5	0.8	0.5	$z-z$	0.39	0.51 ± 0.15	0.40	holds	none
C _r	100	30	3.5	0.8	0.5	$z-z$	0.38	0.83 ± 0.17	1.31	marginal	M1
C _r	100	30	3.5	0.8	0.5	$z-z$	0.36	0.59 ± 0.14	0.83	holds	none
C _r	100	30	3.5	0.8	0.5	$z-z$	0.09	-0.05 ± 0.14	0.51	holds ⁰	none
H	0	15	3.0	0.4	1.0	$z-y$	0.00	0.09 ± 0.08	0.56	holds ⁰	none
H	2.5	15	3.0	0.4	1.0	$z-y$	0.08	0.09 ± 0.08	0.10	holds ⁰	none
H	5	15	3.0	0.4	1.0	$z-y$	0.26	0.10 ± 0.08	1.00	holds	none
H	7.5	15	3.0	0.4	1.0	$z-y$	0.47	0.26 ± 0.08	1.25	marginal	M1/EM-channel
H	10	15	3.0	0.4	1.0	$z-y$	0.67	0.42 ± 0.08	1.48	marginal	M1/EM-channel
H	12.5	15	3.0	0.4	1.0	$z-y$	0.83	0.43 ± 0.09	2.30	breaks	M1/EM-channel
H	15	15	3.0	0.4	1.0	$z-y$	0.96	0.74 ± 0.09	0.95	holds	none
H	17.5	15	3.0	0.4	1.0	$z-y$	1.04	0.77 ± 0.09	1.13	marginal	M1/EM-channel
H	20	15	3.0	0.4	1.0	$z-y$	1.07	0.71 ± 0.09	1.69	marginal	M1/EM-channel
H	22.5	15	3.0	0.4	1.0	$z-y$	1.08	0.79 ± 0.09	1.23	marginal	M1/EM-channel
H	25	15	3.0	0.4	1.0	$z-y$	1.05	0.75 ± 0.09	1.33	marginal	M1/EM-channel
H	27.5	15	3.0	0.4	1.0	$z-y$	1.01	0.73 ± 0.09	1.30	marginal	M1/EM-channel
H	30	15	3.0	0.4	1.0	$z-y$	0.96	0.81 ± 0.09	0.62	holds	none
H	0	15	3.0	0.4	1.0	$z-z$	2.14	2.14 ± 0.11	0.00	holds	none
H	2.5	15	3.0	0.4	1.0	$z-z$	2.11	2.06 ± 0.11	0.08	holds	none
H	5	15	3.0	0.4	1.0	$z-z$	2.00	1.87 ± 0.11	0.23	holds	none
H	7.5	15	3.0	0.4	1.0	$z-z$	1.89	1.76 ± 0.11	0.26	holds	none

	$ B $ (MG)	ϕ_B (deg)	n_e	T_e (keV)	I	pol.	g_{lin}	$g_{\text{PIC}} \pm \sigma$	δ/τ	verdict	mech.
H	10	15	3.0	0.4	1.0	$z-z$	1.76	1.37 ± 0.10	0.93	holds	none
H	12.5	15	3.0	0.4	1.0	$z-z$	1.60	1.46 ± 0.10	0.34	holds	none
H	15	15	3.0	0.4	1.0	$z-z$	1.44	1.12 ± 0.10	0.92	holds	none
H	17.5	15	3.0	0.4	1.0	$z-z$	1.27	1.06 ± 0.09	0.65	holds	none
H	20	15	3.0	0.4	1.0	$z-z$	1.12	1.06 ± 0.09	0.20	holds	none
H	22.5	15	3.0	0.4	1.0	$z-z$	1.01	0.87 ± 0.09	0.53	holds	none
H	25	15	3.0	0.4	1.0	$z-z$	0.95	0.85 ± 0.09	0.40	holds	none
H	27.5	15	3.0	0.4	1.0	$z-z$	0.94	0.71 ± 0.08	1.09	marginal	M1/EM-channel
H	30	15	3.0	0.4	1.0	$z-z$	0.95	0.81 ± 0.08	0.59	holds	none
H	0	15	3.0	1.2	1.0	$z-y$	-0.00	0.05 ± 0.08	0.31	holds ⁰	none
H	2.5	15	3.0	1.2	1.0	$z-y$	0.02	0.03 ± 0.08	0.03	holds ⁰	none
H	5	15	3.0	1.2	1.0	$z-y$	0.08	0.08 ± 0.08	0.02	holds ⁰	none
H	7.5	15	3.0	1.2	1.0	$z-y$	0.16	0.12 ± 0.09	0.23	holds	none
H	10	15	3.0	1.2	1.0	$z-y$	0.24	0.11 ± 0.08	0.77	holds	none
H	12.5	15	3.0	1.2	1.0	$z-y$	0.32	0.13 ± 0.08	1.18	marginal	M1/EM-channel
H	17.5	15	3.0	1.2	1.0	$z-y$	0.43	0.18 ± 0.13	1.00	marginal	M1/EM-channel
H	20	15	3.0	1.2	1.0	$z-y$	0.46	0.21 ± 0.08	1.50	marginal	M1/EM-channel
H	0	15	3.0	1.2	1.0	$z-z$	0.99	0.67 ± 0.12	1.33	marginal	M7
H	2.5	15	3.0	1.2	1.0	$z-z$	0.98	0.47 ± 0.14	1.83	marginal	M7
H	5	15	3.0	1.2	1.0	$z-z$	0.93	0.44 ± 0.15	1.66	marginal	M7
H	7.5	15	3.0	1.2	1.0	$z-z$	0.85	0.50 ± 0.11	1.59	marginal	M7
H	10	15	3.0	1.2	1.0	$z-z$	0.78	0.45 ± 0.16	1.06	marginal	M7
H	12.5	15	3.0	1.2	1.0	$z-z$	0.69	0.52 ± 0.08	1.07	marginal	M7
H	15	15	3.0	1.2	1.0	$z-z$	0.61	0.43 ± 0.08	1.11	marginal	M7
H	17.5	15	3.0	1.2	1.0	$z-z$	0.54	0.30 ± 0.12	1.03	marginal	M7
H	20	15	3.0	1.2	1.0	$z-z$	0.49	0.28 ± 0.09	1.18	marginal	M7